159102 workshop – Week 12

1. Write a C++ program that reads in a name (could be more than one word) and an age. Then display the name and the age. Do NOT include the line `using namespace std;`
Edit the program so that it runs correctly.

2. Write a C++ program that defines three integer variables inside a namespace called RGBvalues. The program must load the values for a particular colour and then display the values.

3. Modify the program in Question 2 to include the line `using namespace RGBvalues;`

4. Write a C++ program that uses the following class along with the global array:

```
class vehicle_class {
private:
  string plate; // e.g. "ABC123"
  int year;     // e.g. 2004
public:
  void loaddata(string p, int y);
  void display();
};

vehicle_class fleet[100];
```

The program must read in details for a number of vehicles (actual number is unknown) and then display the information for every vehicle. You will need to write code for the methods.

5. Modify the program in Question 4 to become a system program that uses command line arguments. All `cin` statements must be removed from the program. You will run the program from the command prompt and you will then enter something that looks like this:

```
prog1 ABC123 2001 GHY869 2015 HDT293 2018
```

Then display the information for every vehicle (as in Question 4). Note that you still do not know how many vehicles will be entered. Your program must calculate how many data entries are coming in from the command line.

6. Do not actually run the program but predict what the following program is going to display. Note: the answer "it will display zero" is not correct.

```
#include <iostream>
using namespace std;

void thing();
```

```
int main() {
  thing();
}

void thing() {
  int value;
  cout << value << endl;
}
```